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Use the Ratio Test to determine if the following series converges absolutely or diverges.

$$\sum_{n=1}^{\infty} \frac{13^n}{n!}$$

Let $\sum a_n$ be any series and suppose that $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \rho$. The Ratio Test states that the series converges absolutely if $\rho < 1$ and diverges if $\rho > 1$ or ρ is infinite. The test is inconclusive if $\rho = 1$.

Let $a_n = \frac{13^n}{n!}$. First, determine $\left| \frac{a_{n+1}}{a_n} \right|$.

$$\begin{aligned} \left| \frac{a_{n+1}}{a_n} \right| &= \left| \frac{\frac{(13^{n+1})}{(n+1)!}}{\frac{(13^n)}{n!}} \right| \\ &= \left| \frac{13^{n+1}}{(n+1)!} \cdot \frac{n!}{13^n} \right| \end{aligned}$$

Simplify the factorials.

$$\begin{aligned} \left| \frac{a_{n+1}}{a_n} \right| &= \left| \frac{13^{n+1}}{13^n} \cdot \frac{n!}{(n+1)!} \right| \\ &= \left| \frac{13^{n+1}}{13^n} \cdot \frac{1}{n+1} \right| \end{aligned}$$

Simplify $\frac{13^{n+1}}{13^n}$.

$$\frac{13^{n+1}}{13^n} = 13$$

Therefore, $\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{13}{n+1} \right|$.

Evaluate $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$.

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{13}{n+1} \right| \\ &= 0 \end{aligned}$$

Since $\rho = 0$, which is less than 1, by the Ratio Test, the series $\sum_{n=1}^{\infty} \frac{13^n}{n!}$ converges absolutely.